

NORTHGATE ACADEMY

Year 10 Combined Science - Biology Paper 1

1 Name the organelle that carries out aerobic respiration, and state the molecule it produces. [2 marks]

Answer:

The mitochondrion. It makes ATP.

2 (a) Write the word equation for aerobic respiration.
[1 mark]

Answer:

glucose + oxygen $\rightarrow$ carbon dioxide + water

2 (b) Explain why muscle cells contain a large number of mitochondria. [2 marks]

Answer:

Because muscle cells need a lot of energy,
especially when you are exercising.

3 A cell is 50 micrometres long. In a drawing the same cell measures 100 mm. Calculate the magnification. [3 marks]

Answer:

50 micrometres = 0.05 mm
magnification = size of image / size of real object
= $100 / 0.05$
= 2000
Magnification = x2000

4 Describe two ways a root hair cell is adapted for taking up mineral ions from the soil. [2 marks]

Answer:

It has a long thin extension that sticks out into the soil, which increases the surface area. It also has lots of mitochondria to provide the energy.

5 State one function of the cell membrane.
[1 mark]

Answer:

7 Define osmosis.
[3 marks]

Answer:

Osmosis is when water moves from a dilute solution to a more concentrated solution.